Remote Qualifier Round
Team Packet
2023 Southwest Regional CCDC

https://southwestccdc.com/

February 14, 2023

Game Time

The 2023 Southwest CCDC Qualifier will be held from 2:00PM—6:00PM Central time on Saturday February 18th. We would suggest your team login to Discord and Portal to troubleshoot any issues well in advance.

Game Organization

The game is divided into several teams, each named with a color. Student competitors are on "blue" teams. Game officials are "gold" team. Gold team is not part of the scenario and perform all judging and game administration. The "white" team operates the simulated business and policy aspects of the game. A team of professional penetration testers and will compose "red" team, with the explicit goal of attacking your network. The "black" team handles network operations and game infrastructure.

This document serves as a supplement to the official National CCDC rules (http://nationalccdc.org/index.php/competition/competitors/rules). Reading this document is not a substitute for reading the rules. All team members will be expected to have read this document and the rules in their entirety. DO NOT JUST SKIM THEM.

On-site Judge

We will not be requiring any judge for this year's qualification round. Team coaches are not allowed in the room(s) students are participating in.

However, competitors **must monitor the Discord for announcements and clarifications**.

Competition Infrastructure

Blue teams will compete using their own computers, which connect to a virtual competition environment using a VPN. The competition environment includes a simulated "Internet", your corporate office, a cloud provider, and some upstream services you and your company consume. We also provide a VirtualBox-compatible virtual appliance pre-configured to connect to the competition environment that you are encouraged, but not required, to use to compete.

Note: blackteam accounts. Many systems will have accounts named "blackteam". These are legitimate competition infrastructure, used for debugging and for scoring injects. The red team is forbidden to impersonate these. Do not remove them or change their credentials.

Simulated Internet

The majority of the private IP network space in the competition environment is a simulated Internet. Red team activity and scoring checks can come from a variety of sources across this network. It also contains some upstream services for your company, which may include DNS, document collaboration, a certificate authority, simulated social media, and/or other services.

The environment includes an Etherpad Lite server, which is available for collaborating on documents and inject responses. It is capable of exporting PDFs that can be uploaded to the Portal as inject responses.

Corporate Networks

Your company has one or more corporate networks in the competition environment. You may be able to access them directly with a VPN, or via console access in our cloud tool, or across the competition simulated Internet from the computer you use to compete from. These networks provide both internal and external services, and depending on the service and its users they may be scored from inside the corporate network, outside the corporate network, or both.

Identify these networks by looking for segments with labels like "office" or "server" in the provided network diagram.

OpenNebula private cloud

In addition to the corporate office (bottom half of network diagram), your company operates a cloud environment (top half of the network diagram). The cloud environment is reachable over the network once you're on the competition VPN. It's also managed via the SWCCDC private cloud (running OpenNebula) at `https://cloud.southwestccdc.com`. You should think of cloud.southwestccdc.com as analogous to, for example, the AWS or Azure web console. Connecting to these instances over the network on the VPN, then, is analogous to connecting to public cloud instances over the Internet. SSHing directly over

the VPN to these servers, or using other remote management tools to access them, is encouraged!

Credentials for the SWCCDC cloud will be sent in the Portal "Announcements" tab. These will be available prior to the competition. Please use them to test your connectivity following the instructions below; however, do not attempt to create or modify any cloud VMs or other resources prior to the competition start.

Portal

Most important pre-game and in-game competition management activities happen on the SWCCDC Portal, located at https://portal.southwestccdc.com. It manages business injects, policy, incident response, and scoring functions. **Note: the SWCCDC Portal logins and passwords are not the same as those for the SWCCDC cloud**.

The portal is open to the Internet, so you can access it without being on the competition network or on VPN, although access from within the environment is allowed. From the portal, you will be able to see your team's service up/down status as seen from the scoring engine. The portal also includes a manual, which is accessible via the "manual" link in the menu and details how to use various features.

Team captains will register first, and then via the "Team" tab invite every team member via Email. If a team member has already signed up before being invited, the captain can add them to the team via inviting the same email the student signed up with.

Coaches are required to have a portal account.

After creating your portal account, you must add your discord ID to your profile. Instructions on how to find your ID can be found in the portal profile tab.

Ports and Firewall info

Students will need HTTP/HTTPS access to portal.southwestccdc.com, cloud.southwestccdc.com, and Discord. Your computer will also need to establish connections with the following destination ports or applications:

```
OpenVPN: 129.244.246.64 port 1195 udp
```

Discord

We will use Discord as the primary way of sending out of game messages, and occasionally in-game messages. Injects will not come from Discord, and no inject submissions, incident response, or other scoring component will be submitted via Discord unless explicitly noted. Meaning, if you upload a response to an inject only to Discord, you will not receive points for it.

You will have a handful of both text and voice channels for your team to use as you see fit during the competition. Please note that other school's coaches are permitted to listen in on your chats and voice conversations. They will not be able to communicate to you, however.

There will also be several help channels. Please ask for help in the dedicated channels and not in per-team channels. We also have a briefing channel where we will meet both before and after the event.

Privacy note

All activity on the competition network is subject to monitoring. It is not our goal to invade your privacy, but we may take measures to observe any network activity to enforce the competition rules, general fairness, and acceptable use of the competition networks. Accordingly, there are some specific privacy considerations when using the competition environment:

VPN. Split tunneling in the blue team VPN(s) *is* allowed, meaning that only traffic destined for CCDC competition networks will flow across the VPN. However, a properly configured VPN connection will set our servers to your DNS resolver. *We are not actively snooping on your DNS queries*, but be aware that this data may leak onto CCDC networks even when the traffic destined for the hosts you resolve may not.

HTTPS. The download package includes a root CA cert for you to install. It comes pre-installed on the Gamestation VM. This will allow in-competition HTTPS servers to offer validating HTTPS connections to clients, signed by a SWCCDC certificate authority. If you install this certificate on any system other than the Gamestation VM (where it is pre-installed), *remove it after the competition*. Real commercial certificate authorities protect their private keys with much stronger measures than we are using. Anyone who compromises this root certificate's private keys will be able to offer valid-looking TLS connections to anyone who still has this certificate trusted after the competition.

Download package

The VPN configurations, optional blue team Gamestation VM appliance, and any other files you may need to download to compete are included in the Download Package. Please download it at least 48 hours before the competition begins. Any updates to documentation will also be included in the download package.

Find the link to the download package immediately below. It is hosted on OneDrive. If it asks for a login, try incognito mode.

⟨https://1drv.ms/u/s!AoFtBT9cDltHgo9nwnxfDPK-ek8psQ?e=FHLIzx⟩—SWCCDC 2023 Qualifier download package.

Gamestation VM

The Gamestation VM is an optional platform for competing in the Southwest Regional Collegiate Cyber Defense Competition Qualifier Round. It is pre-configured with a VPN

that places you in the competition environment, and there are some useful shortcuts on the desktop to sites you'll need to access frequently during the competition. The system runs Xubuntu 22.04 LTS.

This VM is designed to run in VirtualBox. It has been tested on VirtualBox 7.0.6 r155176 on Windows 11 Pro 22H2 22621.1194. Bidirectional clipboard sharing support is enabled and tested as working. However, no file sharing capabilities have been tested, although it should be possible to do so. If you experience strange graphics behaviors, consider disabling the 3D graphics acceleration for the VM in VirtualBox settings. However, the 3D acceleration support is required for UI scaling within Xfce.

It's highly recommended you also install the Oracle VirtualBox Extension Pack for your HOST system (not inside the VM, but on the computer that VirtualBox is installed on).

The login credentials are: `swccdc` : `swccdc`. Passwordless sudo is allowed. Feel free to customize this VM however you want, including before the competition, as long as you don't break the rules by loading custom scripts or applications.

This is a trusted image and is not subject to pre-competition tampering by the red team. Once connected to the competition environment, usage is at your own risk. The default VPN places this machine on a network that is out of scope for the red team, but it does provide connectivity into your teams' in-scope networks.

Competition VPN(s)

You are highly encouraged to use the provided VPNs to interact with the competition. This VM can connect to the competition environment! Use it to SSH to Linux servers, to RDP to Windows servers, to interact with your team's network, and test whether your services appear to be up!

The download package includes the VPN files you'll need prior to the competition. It's possible more may be provided during or shortly before the competition.

Internet VPN

The Internet VPN simulates your a home or public Internet connection in the competition environment. It is outside the control of your company.

This is the default VPN. It will place you into the competition network, but outside of your team's corporate network. Use it to access your company's public services, and to test your services from the perspective of an external customer.

Office VPN

This VPN is not provided prior to the competition, and it is not guaranteed to exist at all.

Any Office VPN will simulate a corporate office network. It places you behind the corporate firewall and enables internal access to your services like you are in the corporate office.

During the competition, you MAY receive this additional VPN configuration. If it has a name like this, it will place you inside your corporate office network as though you were an employee. Take advantage of the different network access between the two VPNs to test your services from different points on the network.

Business Scenario: Edron

Edron is a municipality heavily dependent on energy production and distribution in a world surprisingly similar to, but legally distinct from, the world of Tron. As punishment for your humanistic properties, you have been assigned to securing and maintaining operations of Edron's power distribution offices and cloud environments. Failure to comply with Edron policy and business requirements comes with dire consequences.

For centuries former "employees," no longer in this world, have assembled and left behind the following documents to provide their successors the opportunity for a more human ending to their Edron career.

Scenario Network Info

This diagram was found on the floor in the office and may be incomplete or inaccurate, but it's what you have:

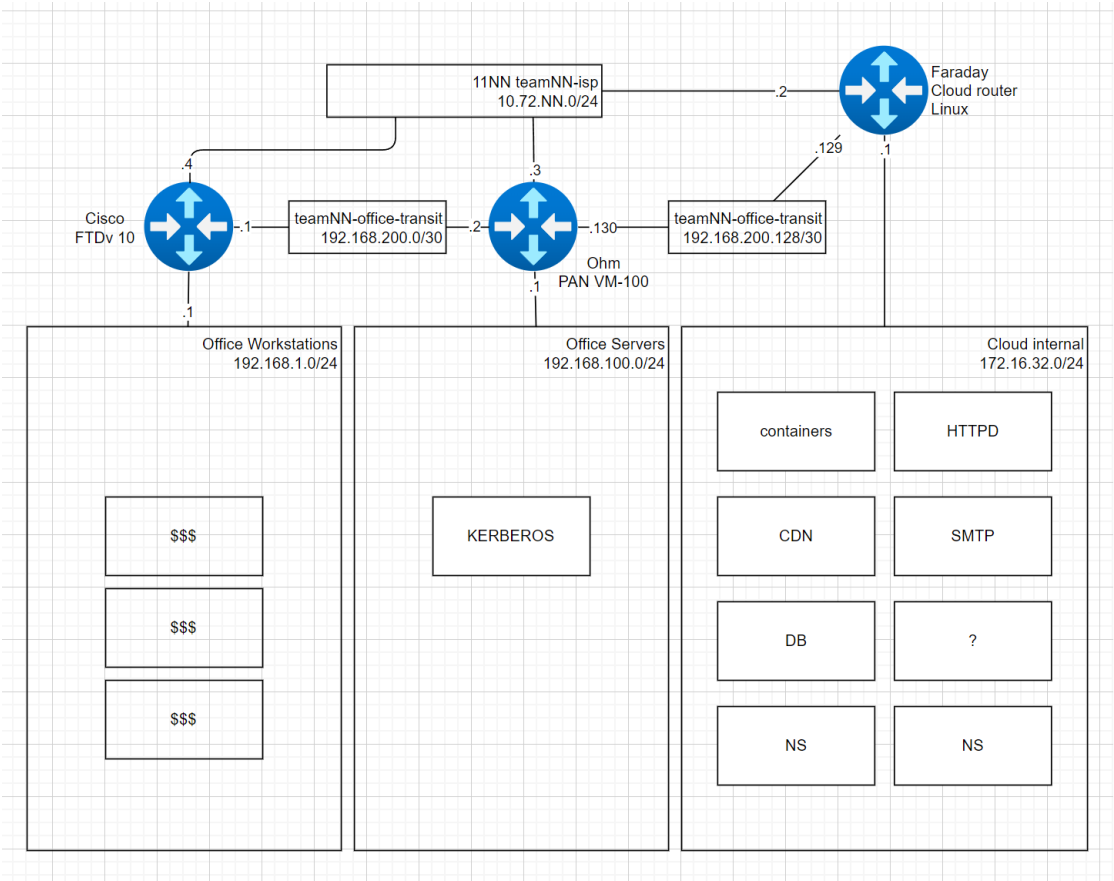

Figure 1: Edron Scenario Network Overview

Connection test

Blue teams should test their connectivity to the competition environment prior to the competition. Connection testing is available from the release of the connection details until 24 hours before the competition begins.

Preparing to test the Gamestation VM

The Gamestation VM is ready to connect to the competition environment upon launch, assuming it has a working Internet connection.

Preparing to test your own computer

For connecting from a system other than the Gamestation VM, you will need to install the blue team Internet VPN. We use OpenVPN for this. The VPN files are provided in the download package, and they include general instructions on their use. We also provide a self-executing Windows installer that includes both the OpenVPN software and the appropriate configuration.

For HTTPS connections to validate properly inside the environment, you will also need to install the SWCCDC 2023 CA root certificate provided in the download package. Installing this is optional, but may be desireable to avoid TLS errors inside the competition environment. Certificate installation methods vary by platform and browser, so we are unable to provide specific step by step instructions for this process.

Cloud interface test

First, test that you can connect to `https://cloud.southwestccdc.com`. This is our OpenNebula private cloud interface. If you can reach this page, your connectivity to the cloud over the Internet works fine.

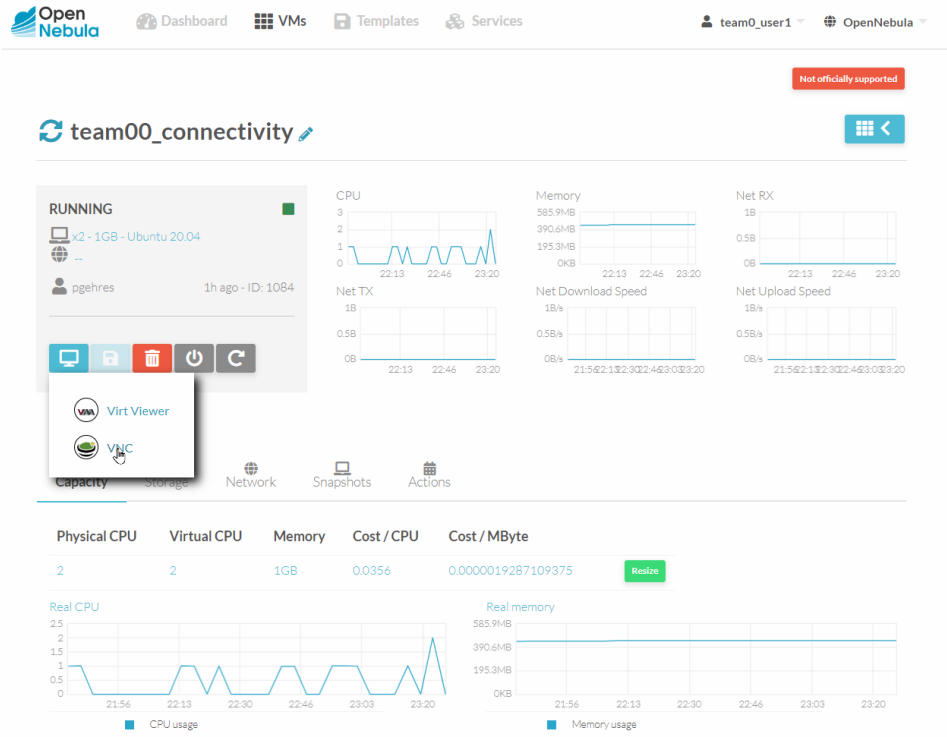

Figure 2: Opening a virtual console in OpenNebula

Please collect your SWCCDC Cloud credentials from the Portal's Announcements tab.

Use them to log into the cloud interface. Test that you can use the cloud interface properly by clicking the "VMs" tab at the top of the page, selecting the teamNN_connectivity VM (where NN is replaced by your team number) by clicking on its name. Then click the monitor icon and select "VNC." See the screenshot in Fig. 2 for where to click. You should then see a Linux login screen. You won't be able to log in to this server, but you should be able to interact with it. If so, your cloud interface test has succeeded.

Internal connectivity test

To test connectivity within the competition environment, make sure you're connected to the Blue Team Internet VPN. Navigate to https://etherpad.edron.swccdc.com. If the Etherpad landing page loads, your connectivity is good! If there are no HTTPS certificate errors, this means you have also successfully installed the SWCCDC root certificate.

You're ready to compete!

Scoring

Teams accumulate points through maintaining functional network services and by corresponding with the simulated business environment through documents called injects. Injects can be documented technical tasks or business and policy focused. Injects and Services each account for approximately half of the total points available to teams. Red Team activity deducts points accumulated from injects and services scores.

NOTE: This qualification event we will be using DNS for scoring services. The scoring engine will use the IP your DNS servers provide for each scored service when preforming a check.

Services

Teams will be able to see service status to monitor the operational status of their network. Services are checked on random intervals every few minutes. Each of these service checks are worth points. If a service is down, it will not grant any points to the team. All services will be functional when you gain access to the network. It is the team's responsibility to secure them and ensure their continued operation. You will be provided a list of services you are operating along with the network diagram a few days before the competition.

Business Injects

Teams will be responsible for answering requests, memos and correspondence in a professional manner. Take care to understand what each request is asking for in detail. Do not provide a short memo when asked for a report and do not provide a report as a paragraph. All responses should include the team number, not a school name. Responses to injects and requests for policy related items are scored heavily and can influence the

final outcome of the game. Late responses are penalized heavily once they are past the due date, which will be by the minute. A memo reply requested by 10:00 AM that is received at 10:01 will be penalized no less than 50-percent immediately before any subject-matter scoring is completed.

Injects can be scored in a variety of ways. For example, if an inject asks you to enable a service on a host, the inject might be worth 0 points however the scoring can come through a scoring engine checking the new service. Don't assume that injects worth low or 0 points are not important.

Red Team Engagement

Red team engagement will occur as soon as the game begins. Red team will not have access to this document unless you leak it to them. Aside from gaining login access, the Red Team may also score points by gaining access to confidential data, such as trade secrets, personally identifiable employee or customer data, or other sensitive materials. You have the opportunity to submit an incident response memo whenever you have been compromised. A detailed description of what machine you believe was compromised, it's IP address, the time-frame you believe the attack occurred and what can be done to mitigate future similar attacks is requested. A memo that reads "someone scanned us" will be disregarded. Accurate incident responses can recover up to half of the points lost from Red-Team activity. If there was no documented attack or access by red team, you will not regain any points.

NOTE: Red team has been advised not to pursue "scorched-earth" style attacks. If your machine doesn't boot and the partition table is gone, it was likely a blue-team action.

NOTE: DO NOT REPORT PORT SCANS. SCANS ARE NOT A COMPROMISE.

Post-Game

The top eight teams will advance to the regional game in March. We will send out the list of teams advancing within a day or two of the qualifier. Teams will be provided with team-specific feedback from qualifiers within 1-2 weeks. There are typically around 20 teams participating in the qualifier and compiling feedback takes some time. Please be patient after the game for specific feedback. Details about how and whether you will be able to access your qualifiers machines after the game ends will be shared during the competition.

Resource Links

- Full Game Rules @ National CCDC
- Mubix's How to win CCDC GitHub

- National CCDC Team Prep Guide
- SWCCDC - "What is CCDC" video on YouTube (15 min)
- Intro to CCDC and 2018 scenario review video on YouTube
- Intro to CCDC and 2018 scenario review slides (PDF)